URDF Model

1. URDF Model Introduction

The Unified Robot Description Format (URDF) is an XML file format widely used in ROS (Robot Operating System) to comprehensively describe all components of a robot.

Robots are typically composed of multiple links and joints. A link is defined as a rigid object with certain physical properties, while a joint connects two links and constrains their relative motion.

By connecting links with joints and imposing motion restrictions, a kinematic model is formed. The URDF file specifies the relationships between joints and links, their inertial properties, geometric characteristics, and collision models.

2. Comparison between Xacro and URDF Models

The URDF model serves as a description file for simple robot models, offering a clear and easily understandable structure. However, when it comes to describing complex robot structures, using URDF alone can result in lengthy and unclear descriptions.

To address this limitation, the xacro model extends the capabilities of URDF while maintaining its core features. The xacro format provides a more advanced approach to describe robot structures. It greatly improves code reusability and helps avoid excessive description length.

For instance, when describing the two legs of a humanoid robot, the URDF model would require separate descriptions for each leg. On the other hand, the xacro model allows for describing a single leg and reusing that description for the other leg, resulting in a more concise and efficient representation.

3. Basic Syntax of URDF Model

3.1 XML Basic Syntax

The URDF model is written using XML standard.

Elements:

An element can be defined as desired using the following formula:

<element>

</element>

Properties:

Properties are included within elements to define characteristics and parameters. Please refer to the following formula to define an element with properties:

<element property_1="property value1" property_2="property value2">

</element>

Comments:

Comments have no impact on the definition of other properties and elements. Please use the following formula to define a comment:

<!-- comment content -->

3.2 Link

The Link element describes the visual and physical properties of the robot's rigid component. The following tags are commonly used to define the motion of a link:

```
<link name="<link name>">
    <inertial>......</inertial>
    <visual>......</visual>
    <collision>......</collision>
</link>
```

<visual>: Describe the appearance of the link, such as size, color and shape.

<inertial>: Describe the inertia parameters of the link, which will used in dynamics calculation.

<collision>: Describe the collision inertia property of the link

Each tag contains the corresponding child tag. The functions of the tags are listed below.

Tag	Function
origin	Describe the pose of the link. It contains two parameters, including xyz and rpy. Xyz describes the pose of the link in the simulated map. Rpy describes the pose of the link in the simulated map.
mess	Describe the mess of the link
inertia	Describe the inertia of the link. As the inertia matrix is symmetrical, these six parameters need to be input, ixx, ixy, ixz, iyy, iyz and izz, as properties. These parameters can be calculated.
geometry	Describe the shape of the link. It uses mesh parameter to load texture file, and em[ploys filename parameters to load the path for texture file. It has three child tags, namely box, cylinder and sphere.
material	Describe the material of the link. The parameter name is the required filed. The tag color can be used to change the color and transparency of the link.

3.3 Joint

The "**Joint**" tag describes the kinematic and dynamic properties of the robot's joints, including the joint's range of motion, target positions, and speed limitations. In terms of motion style, joints can be categorized into six types.

Type and Explanation	Tag
continuous joint: rotate around single axis continuously	continuous
revolute joint: similar to continuous, but its rotation angle is limited	revolute
prismatic joint: move along a axis within limited range	prismatic
Planar joints: translate or rotate in plane orthogonal directions	planar
floating joint: translate and rotate	floating
fixed joint: not allowed to do any movements	fixed

The following tags will be used to write joint motion.

```
<joint name="name of joint">
    <parent link="link1">
    <child link="link2">
    <calibration>......</calibration>
    <dynamics damping ....../>
    <limit effort ....../>
</joint>
```

<parent_link>: Parent link

<child_link>: Child link

<calibration>: Calibrate the joint angle

<dynamics>: Describes some physical properties of motion

<limit>: Describes some limitations of the motion

The function of each tag is listed below. Each tag involves one or several child tags.

Tag	Function
origin	Describe the pose of the parent link. It

	involves two parameters, including xyz and rpy. Both xyz and rpy describe the pose of the link in simulated map.
axis	Control the child link to rotate around any axis of the parent link.
limit	The motion of the child link is constrained using the lower and upper properties, which define the limits of rotation for the child link. The effort properties restrict the allowable force range applied during rotation (values: positive and negative; units: N). The velocity properties confine the rotational speed, measured in meters per second (m/s).
mimic	Describe the relationship between joints.
safety_controller	Describes the parameters of the safety controller used for protecting the joint motion of the robot.

3.4 Robot Tag

The complete top tags of a robot, including the <link> and <joint> tags, must be enclosed within the <robot> tag. The format is as follows:

```
<robot name="name of robot">
    <link>......</link>
    <link>......</link>

    <joint>......</joint>
    <joint>......</joint>
</robot>
```

3.5 gazebo Tag

This tag is used in conjunction with the Gazebo simulator. Within this tag, you can define simulation parameters and import Gazebo plugins, as well as specify Gazebo's physical properties, and more.

```
<gazebo reference="link1">
    <material>Gazebo/Black</material>
</gazebo>
```

3.6 Write Simple URDF Model

Name the model of the robot

To start writing the URDF model, we need to set the name of the robot following this format: “**<robot name=“robot model name”>**”. Lastly, input “**</robot>**” at the end to represent that the model is written successfully.

```
<?xml version="1.0"?>
<robot name="pan_tilt">
```

```
</robot>
```

Set links

1) To write the first link and use indentation to indicate that it is part of the currently set model. Set the name of the link using the following format: **<link name="link name">**. Finally, conclude with "**</link>**" to indicate the successful completion of the link definition.

```
        <link name="base_link">
```

```
        </link>
```

2) Write the link description and use indentation to indicate that it is part of the currently set link, and conclude with "</visual>".

```
                <visual>
```

```
                <visual>
```

3) The "**<geometry>**" tag is employed to define the shape of a link. Once the description is complete, include "**</geometry>**". Within the "**<geometry>**" tag, indentation is used to specify the detailed description of the link's shape. The

following example demonstrates a link with a cylindrical shape: "**<cylinder length="0.01" radius="0.2"/>**". In this instance, "**length="0.01"**" signifies a length of 0.01 meters for the link, while "**radius="0.2"**" denotes a radius of 0.2 meters, resulting in a cylindrical shape.

```
			<geometry>
				<cylinder length="0.01" radius="0.2" />
			</geometry>
```

4) The "**<origin>**" tag is utilized to specify the position of a link, with indentation used to indicate the detailed description of the link's position. The following example demonstrates the position of a link: "**<origin rpy="0 0 0" xyz="0 0 0" />**". In this example, "**rpy**" represents the roll, pitch, and yaw angles of the link, while "**xyz**" represents the coordinates of the link's position. This particular example indicates that the link is positioned at the origin of the coordinate system.

```
			<origin rpy="0 0 0" xyz="0 0 0" />
```

5) The "**<material>**" tag is used to define the visual appearance of a link, with indentation used to specify the detailed description of the link's color. To start describing the color, include "**<material>**", and end with "**</material>**" when the description is complete. The following example demonstrates setting a link color to yellow: "**<color rgba="1 1 0 1" />**". In this example, "**rgba="1 1 0 1"**" represents the color threshold for achieving a yellow color.

```
			<material name="yellow">
				<color rgba="1 1 0 1" />
			</material>
```

Set joint

1) To write the first joint, use indentation to indicate that the joint belongs to the current model being set. Then, specify the name and type of the joint as follows: "**<joint name="joint name" type="joint type">**". Finally, include "**</joint>**" to indicate the completion of the joint definition.

Note: to learn about the type of the joint, please refer to "3.3 Joint".

```
	<joint name="pan_joint" type="revolute">
```

```
</joint>
```

2) Write the description section for the connection between the link and the joint. Use indentation to indicate that it is part of the currently defined joint. The parent parameter and child parameter should be set using the following format: "**<parent link="parent link"/>**", and "**<child link="child link" />**". With the parent link serving as the pivot, the joint rotates the child link.

```
<parent link="base_link" />
<child link="pan_link" />
```

3) “**<origin>**” describes the position of the joint using indention. This example describes the position of the joint: “**<origin xyz=“0 0 0.1” />**”. xyz is the coordinate of the joint.

```
<origin xyz="0 0 0.1" />
```

4) “**<axis>**” describes the position of the joint adopting indention. “**<axis xyz=“0 0 1” />**” describes one posture of a joint. xyz specifies the pose of the joint.

```
<axis xyz="0 0 1" />
```

5) “**<limit>**” imposes restrictions on the joint using indention. The below picture The "**<limit>**" tag is used to restrict the motion of a joint, with indentation indicating the specific description of the joint angle limitations. The following example describes a joint with a maximum force limit of 300 Newtons, an upper limit of 3.14 radians, and a lower limit of -3.14 radians. The settings are defined as follows: "**effort=“joint force (N)”, velocity=“joint motion speed”, lower=“lower limit in radians”, upper=“upper limit in radians**”.

```
<limit effort="300" velocity="0.1" lower="-3.14" upper="3.14" />
```

6) “**<dynamics>**” describes the dynamics of the joint using indention. “**<dynamics damping=“50” friction=“1” />**” describes dynamics parameters of a joint.

```
<dynamics damping="50" friction="1" />
```

The complete codes are as below.

```xml
<?xml version="1.0"?>
<robot name="pan_tilt">

	<!--定义了base_link -->
	<!-- <visual>标签描述了在仿真环境的外观，包括几何外形<geometry>（圆柱形cylinder）等-->
	<link name="base_link">
		<visual>
			<geometry>
				<cylinder length="0.01" radius="0.2" />
			</geometry>
			<origin rpy="0 0 0" xyz="0 0 0" />
			<material name="yellow">
				<color rgba="1 1 0 1" />
			</material>
		</visual>
	</link>

	<!-- 定义了关节pan_joint，以及其关节类型：旋转副（有限制） -->
	<!-- 旋转副连接的两个刚体分别为base_link和pan_link -->
	<joint name="pan_joint" type="revolute">
		<parent link="base_link" />
		<child link="pan_link" />
		<origin xyz="0 0 0.1" />
		<axis xyz="0 0 1" />
		<limit effort="300" velocity="0.1" lower="-3.14" upper="3.14" />
		<dynamics damping="50" friction="1" />
	</joint>

	<link name="pan_link">
		<visual>
			<geometry>
				<cylinder length="0.4" radius="0.04" />
			</geometry>
			<origin rpy="0 0 0" xyz="0 0 0.09" />
			<material name="red">
				<color rgba="0 0 1 1" />
			</material>
		</visual>
	</link>

	<joint name="tilt_joint" type="continuous">
		<parent link="pan_link" />
		<child link="tilt_link" />
		<origin xyz="0 0 0.2" />
		<axis xyz="0 1 0" />
		<limit effort="300" velocity="0.1" lower="-4.64" upper="-1.5"/>
		<dynamics damping="50" friction="1"/>
	</joint>

	<link name="tilt_link">
		<visual>
			<geometry>
				<cylinder length="0.4" radius="0.04" />
			</geometry>
			<origin rpy="0 1.5 0" xyz="0 0 0" />
			<material name="green">
				<color rgba="1 0 0 1" />
			</material>
		</visual>
	</link>

</robot>
```